

CS121P Introduction to Computer Programming Test #1

This Test is open-book (course textbook) and open-notes (your own notes from this class) and open-laptop, but closed-neighbor. You may not communicate with anyone during the Test, other than the Instructor. There are a total of 100 points possible.

Relax, take a deep breath, and pray ☺.

Your grade will be based upon both the correctness and quality of your solutions. Don't get stuck on one portion of the test, but look all of it over first. If you cannot completely solve something, do your best, and submit whatever you are able to, so that I will have more opportunity to award any applicable partial credit.

DO NOT LEAVE THE ROOM until I verify to you that your submission has been received successfully.

For the IN CLASS PORTION of this Test, implement solutions to ANY TWO (your choice) of the following problems. If you choose to implement solutions to more than two of these problems, you will receive normal credit for the two you do the best, plus a little bit of extra credit for each additional one of sufficient quality.

For the TAKE HOME PORTION of this Test, implement solutions to ANY THREE of the following problems.

(continued on next page)

Consider the following source code file “Test1.py”, available for download from Canvas, which successfully implements Homework 04:

```
import turtle
import random

background_red = 0.25 * random.randrange(0, 2)
background_green = 0.25 * random.randrange(0, 2)
background_blue = 0.25 * random.randrange(0, 2)
print("Random background color is: (", background_red, ",",
      background_green, ",", background_blue, ")")

wn = turtle.Screen()
wn.bgcolor(background_red, background_green, background_blue)

rotational_copies = int(input("Enter rotational copies: "))
sides_per_polygon = int(input("Enter sides_per_polygon per polygon: "))
edge_pixel_length = int(input("Enter edge pixel length: "))

murtle = turtle.Turtle()
murtle.hideturtle()
murtle.speed(0)

murtle.pensize(5)
murtle.color(1,1,1)
for i in range(rotational_copies):
    for j in range(sides_per_polygon):
        murtle.forward(edge_pixel_length)
        murtle.left(360/sides_per_polygon)
    murtle.left(360/rotational_copies)

murtle.pensize(1)
murtle.color(0,0,0)
for i in range(rotational_copies):
    for j in range(sides_per_polygon):
        murtle.forward(edge_pixel_length)
        murtle.left(360/sides_per_polygon)
    murtle.left(360/rotational_copies)

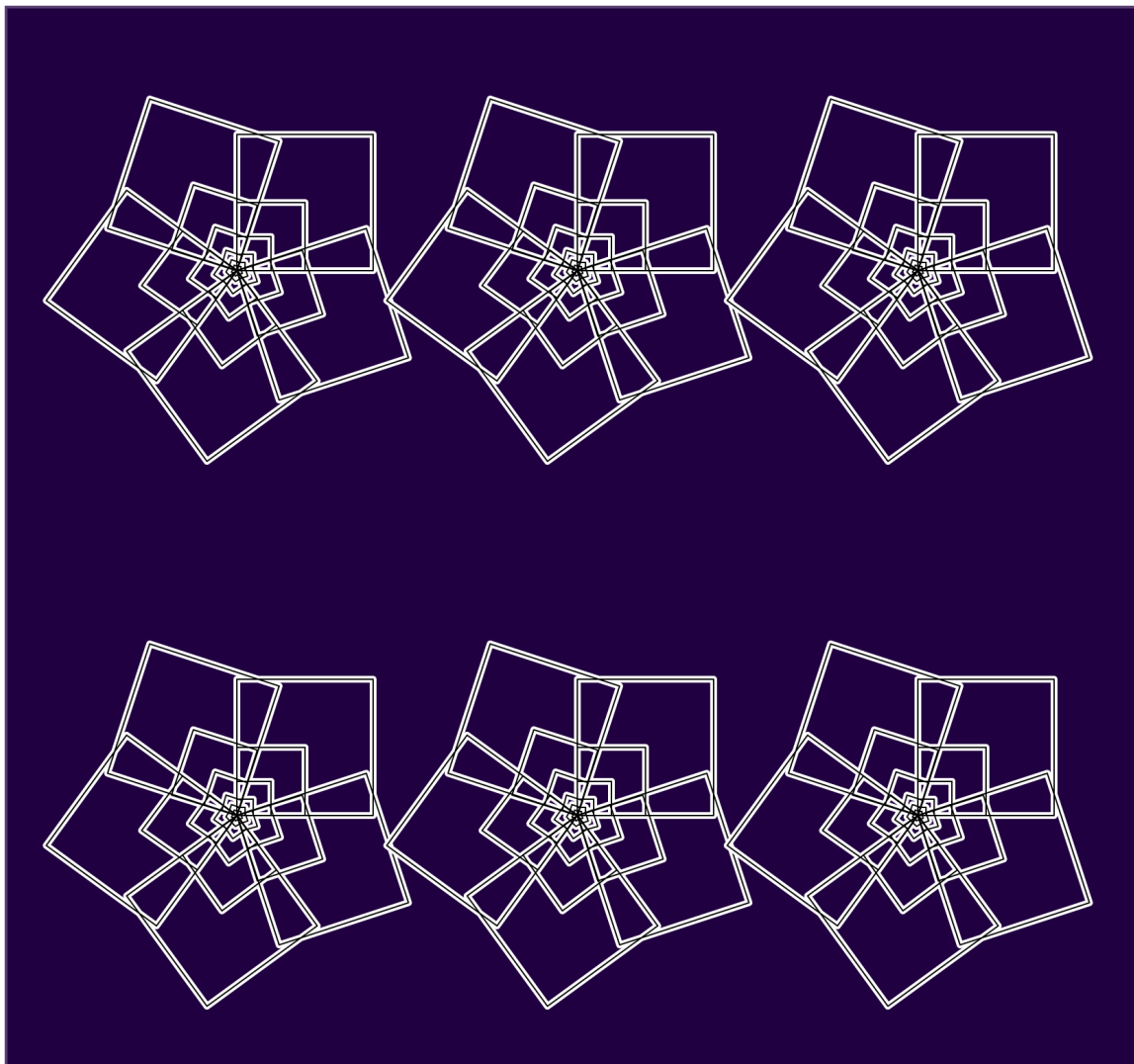
print("Click turtle screen to exit...")
wn.exitonclick()
```

Choose any two of the following modifications for *Test 1 In Class*:

1. Change the background color so that each randomly chosen red, green and blue value is either 0.0, 0.125, 0.25, 0.375, or 0.5. In addition, ensure that black did not end up being the randomly chosen color.
2. The code provided does not include any user-defined functions to better structure the program. Reorganize the code to make use of suitable user-defined functions “**draw_polygon**”, “**draw_rotational_polygons**”, and “**main**”.
3. In addition to each full-sized copy of each figure, draw progressively smaller half-sized copies, in a similar manner to that shown in the sample input/output and screenshot.
4. Instead of drawing only one figure, draw a grid of identical figures, with row and column locations specified and drawn in a similar manner to that shown in the sample input/output and screenshot.

(Note: **Choose any three** of the above modifications for *Test 1 Take Home*.)

Random background color is: (0.125 , 0.0 , 0.25)
Enter rotational copies: 5
Enter sides per polygon: 4
Enter edge pixel length: 100
Enter row range start: -200
Enter row range end: 200
Enter row range increment: 400
Enter col range start: -250
Enter col range end: 250
Enter col range increment: 250
Click turtle screen to exit...



~~~ *END OF CS121P TEST #1* ~~~